UGP Summary Report:
Multiscale Computational Prediction of 3D Macroscopic Electrical Resistivity in Sn-Bi Solder Joints

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Abstract

This report details the computational framework developed to predict the macroscopic electrical resistivity of near-eutectic Sn-Bi solder joints. Traditional geometric approximations frequently fail to account for the localized electrical bottlenecks caused by impurities kinetically trapped at phase boundaries during rapid cooling. To resolve this, a multiscale numerical pipeline was constructed to explicitly simulate the physics of the joint. The model stochastically generates phase segregation, computes transient thermodynamic diffusion of impurities via spatial convolution, maps lattice defect scattering using a modified Nordheim's Rule, and evaluates steady-state conduction across a 10^6-node finite difference resistor network. The model yielded a 3D-corrected effective resistivity prediction of 53.20 $\mu\Omega$·cm, aligning strongly with empirical literature.

1 Computational Methodology

1.1 Stage 1: Microstructure Generation (Kinetic Monte Carlo)

The phase morphology was generated using an Ising-model Kinetic Monte Carlo (KMC) engine on a 1000×1000 discretized spatial grid.

- **Volume Fraction Initialization:** While the Sn-Bi eutectic point is defined as 42/58 by weight percent, the 2D spatial matrix requires volumetric ratios. Accounting for the higher density of Bismuth, the randomized liquid melt was seeded at a physically accurate 49% Sn to 51% Bi spatial volume fraction.

- **Hamiltonian Energy Mechanics:** For each stochastic pixel swap, the local interaction energy was evaluated based on the 8 nearest neighbors. The change in energy, ΔE, was calculated using the normalized interfacial penalty $J = 1.0$:

$$\Delta E = E_{final} - E_{initial} \tag{1}$$

- **Metropolis Acceptance Probability:** Transitions reducing system energy ($\Delta E \leq 0$) were accepted unconditionally. Transitions increasing energy ($\Delta E > 0$) were subjected to a thermodynamic probability threshold:

$$P(\Delta E) = \exp\left(-\frac{\Delta E}{KT}\right) \tag{2}$$

  With the thermal fluctuation scale calibrated to $KT = 0.65$, highly unfavorable thermodynamic shifts result in $P(\Delta E) \to 0$. To maximize computational efficiency across 5×10^7 node evaluations, these probabilities were pre-computed into an exponential lookup table (Table 1). The algorithm terminated after 50 sweeps, representing robust topological convergence and Ostwald ripening.

Energy Change (ΔE)	**Acceptance Probability** $P(\Delta E)$
0	1.0000
1	0.2148
2	0.0461
3	0.0099
4	0.0021
5	0.0005
6	0.0001
7	< 0.0001
8	< 0.0001

Table 1: Pre-computed Metropolis acceptance probabilities for positive energy shifts at $KT = 0.65$.

1.2 Stage 2: Transient Fickian Diffusion

Fick's Second Law ($\frac{\partial C}{\partial t} = D\nabla^2 C$) was explicitly evaluated to model the kinetic trapping of solute atoms resulting from the rapid thermal quench from 138°C to room temperature.

- **Interfacial Detection via Convolution:** To avoid computationally expensive iterative neighbor-checking, phase boundaries (`sn_edges` and `bi_edges`) were isolated using a discrete 2D Laplacian convolution kernel:

$$\nabla^2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \tag{3}$$

  By convoluting this kernel over the boolean phase masks, bulk pixels strictly evaluated to 0, while interfacial nodes evaluated to negative integers (< 0). These exact spatial coordinates were then pinned via Dirichlet conditions based on empirical solvus limits (21% Bi in Sn; 2% Sn in Bi).

- **Grid Mobilities & Physical Asymmetry:** The non-dimensional grid mobilities were set to $D_{Bi} = 0.15$ and $D_{Sn} = 0.10$. This ratio mathematically enforces the reality that Bi diffuses faster into the open tetragonal Sn lattice than Sn diffuses into the dense rhombohedral Bi lattice.

- **Stability limit:** The diffusion ran for exactly 150 time steps (representing the thermal quench window). The mobilities and time step ($dt = 0.1$) were chosen so the stability coefficient satisfied the Courant-Friedrichs-Lewy (CFL) limit ($\frac{D \cdot dt}{\Delta x^2} = 0.015 \ll 0.25$), preventing numerical explosion.

1.3 Stage 3: Lattice Defect Scattering

The continuous concentration gradients were mapped to localized electrical resistance penalties using a modified Nordheim's Rule:

$$\rho(x, y) = \rho_{base} + k \cdot \max(0,\ C(x, y) - C_{eq}) \tag{4}$$

- **Sn Matrix Constants:** Baseline $\rho = 18.0\ \mu\Omega{\cdot}\text{cm}$, scattering coefficient $k = 20.0$.
- **Bi Phase Constants:** Baseline $\rho = 129.0\ \mu\Omega{\cdot}\text{cm}$, scattering coefficient $k = 10,000.0$. This severe scattering multiplier reflects the catastrophic Fermi surface disruption of the semimetal Bismuth lattice when subjected to Tin impurity contamination.

1.4 Stage 4: Macroscopic Electrical Transport Solver

Laplace's equation for steady-state conduction ($\nabla \cdot (\sigma \nabla V) = 0$) was resolved globally by balancing Kirchhoff's Current Law across all adjacent nodes:

$$\sum_{j \in \text{neighbors}} G_{i,j}(V_i - V_j) = 0 \tag{5}$$

- **Finite Difference Formulation:** Because the network was built on a normalized square mesh where internodal length (Δx) and cross-sectional area ($\Delta y \cdot t$) equal unity, the geometric scaling factor canceled out ($R = \rho \frac{l}{A} = \rho \frac{1}{1}$), allowing local internodal resistance R to be numerically identical to local average resistivity ρ.
- **Interfacial Penalty (45.0 $\mu\Omega{\cdot}$cm):** Nodes bridging dissimilar phases received an additional lumped penalty of 45.0 $\mu\Omega{\cdot}$cm. Because a continuum grid cannot natively resolve quantum-level electron tunneling, this value serves as a phenomenologically calibrated parameter, accounting for the contact resistance caused by the Fermi surface mismatch between tetragonal Sn and rhombohedral Bi.
- **Boundary Enforcement:** Dirichlet boundary conditions were enforced by explicitly replacing the corresponding matrix rows, setting the diagonal entries to unity and the right-hand side to the prescribed voltage values. Although this approach introduces localized asymmetry, the Conjugate Gradient solver demonstrated stable convergence for the system. The top and bottom edges utilized zero-flux Neumann conditions by intentionally omitting exterior conductance pathways, physically modeling infinite resistance boundaries.
- **Linear Solver:** The $10^6 \times 10^6$ Compressed Sparse Row (CSR) matrix ($Ax = b$) was resolved iteratively via the Conjugate Gradient (CG) method until residual errors achieved mathematical convergence ($rtol = 1e - 8$).

1.5 Dimensionality Correction (Bakker's 1997 Extrapolation)

To account for planar spatial confinement, which artificially forces electrons through boundary bottlenecks, the raw 2D simulation output was extrapolated using K. Bakker's 1997 Effective Medium Theory. Based on the calculated phase conductivity ratio (λ_D/λ_M) and the Bi volume fraction (c_D), an intermediate parameter X was derived:

$$X = 1 - \left(1 - \frac{\lambda_D}{\lambda_M}\right) c_D \tag{6}$$

Using an empirical shape factor of $r \approx 1.83$, the 3D-corrected relative conductivity (f_{3D}) was calculated from the 2D solver output (f_{2D}) to mathematically simulate Z-axis percolation:

$$f_{3D} = X - \frac{X - f_{2D}}{r} \tag{7}$$

2 Software Infrastructure

The pipeline was constructed using core numerical Python libraries: **NumPy** for vectorized Boolean phase mapping, kernel logic, and Nordheim array arithmetic; **Numba (`@njit`)** for JIT compilation of the KMC engine, bridging Python's iterative overhead to enable $\sim$10-second stabilization of 50 million node transitions; **SciPy (`ndimage.convolve`)** for discrete Laplacian filtering; and **SciPy (`sparse.linalg.cg`)** for resolving the 10^{12}-element sparse Kirchhoff network.

3 Results and Conclusion

The multiscale model successfully mapped quantum-level lattice disruptions into a macroscopic transport framework. The raw 2D Finite Difference solver yielded a planar resistivity of **72.40 $\mu\Omega\cdot$cm**. After applying Bakker's 3D geometric relaxation, the model output a dimensionally corrected effective resistivity of **53.20 $\mu\Omega\cdot$cm**. This accurately aligns with the established physical baseline range (35 to 60 $\mu\Omega\cdot$cm) documented in empirical literature, validating the integrated computational methodology.